A Head Camera Is (Almost) All You Need

An electronics-free UMI glove labelled and observed by one moving camera

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Why a head camera only

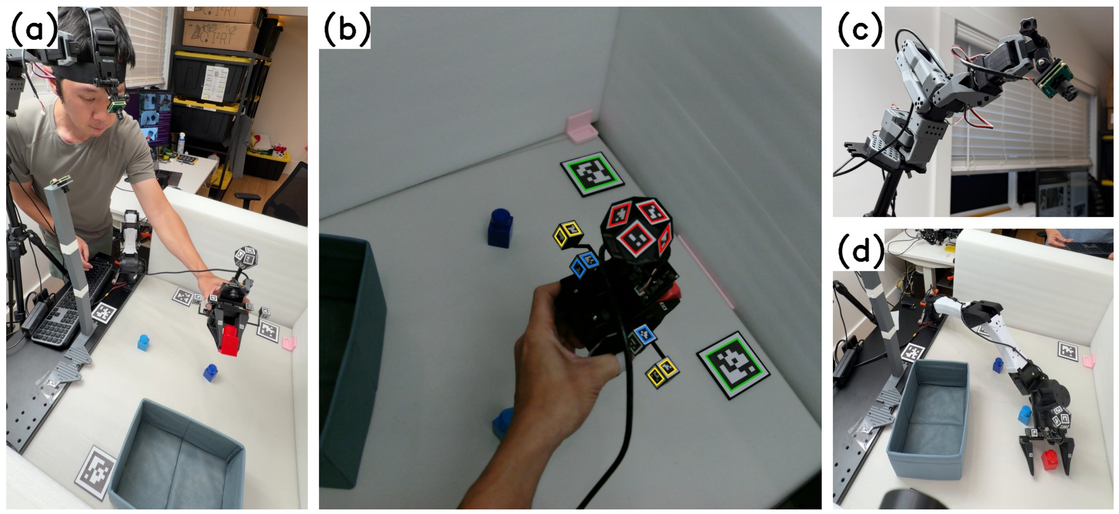
- Wrist devices need a **second stream** to track the gripper: SLAM, an IMU, a controller or an external camera.
- **Gaze leads the hand**: the gripper stays in view, at arm's length, with the goal.
- The frames the policy sees are the frames that **carry the labels**.

No SLAM, no IMU, no wrist camera, no cables.

$144 glove, camera, strap

103 g on the head

One head camera: the policy's observation and the pose labels



(a) A $122 4K head camera, a $22 electronics-free glove, four paper tiles. (b) The camera's own frame: tiles give the camera pose, the ball the gripper pose, the jaw tags the opening. (c) At deployment the same camera rides a $122 six-joint arm (d) beside the robot.

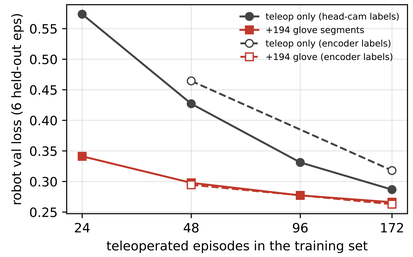
194 glove demos stretch a small teleop set

Eight ACT policies, joint-space actions. Rollouts: 25 placements, one trial each, 60 s. Every co-trained mix uses the same 194 glove demos.

10/25 48 teleop episodes

17/25 48 teleop + 194 glove

1 : 2–3 teleop episode : glove demos



9 placements won to 2 lost ($p = 0.065$); loss 0.464 → 0.295, the level of 96–172 teleop episodes. On the teleop-only curve the glove set is worth **65–100 teleop episodes**, an exchange rate of 2–3 glove demos per teleoperated one. Added to all 172 it still helps: 18/25 → 20/25, loss 0.318 → 0.263.

Head-camera labels can replace the encoders

teleop labelled by	encoders	head camera
validation loss	0.318	0.328 (+3 %)
rollouts	18/25	15/25 ($p = 0.58$)

Same 172 teleop episodes, same held-out encoder targets; 5 placements won to 8 lost.

Label accuracy

- **Camera pose** from the paper tiles, refit from the footage to 0.5 px.
- **Gripper pose** by PnP on a 15 mm ArUco dodecahedron.
- **Jaw width** from printed tails on the jaws.

6.5 mm tool point **1.6°** orientation **3.5 %** jaw width, of stroke

Median error against the robot's encoders; a label on 93–100 % of frames.

Collection

2–3× demos per minute vs teleop **198** demos in four sessions

6–8 s per demo; teleop 14 s **0** robot time

Continuous recording, episodes cut by a keypress; 0.7–7.4 % of frames dropped for tracking.

Deployment

- The same camera on a $122 six-joint arm beside the YAM.
- It follows the gripper **from the robot's encoders alone**, in teleoperation and in rollouts, so the policy sees what the collector saw.

46 mm head pose vs the collector **5.4°** static camera: 87 mm, 7.0°

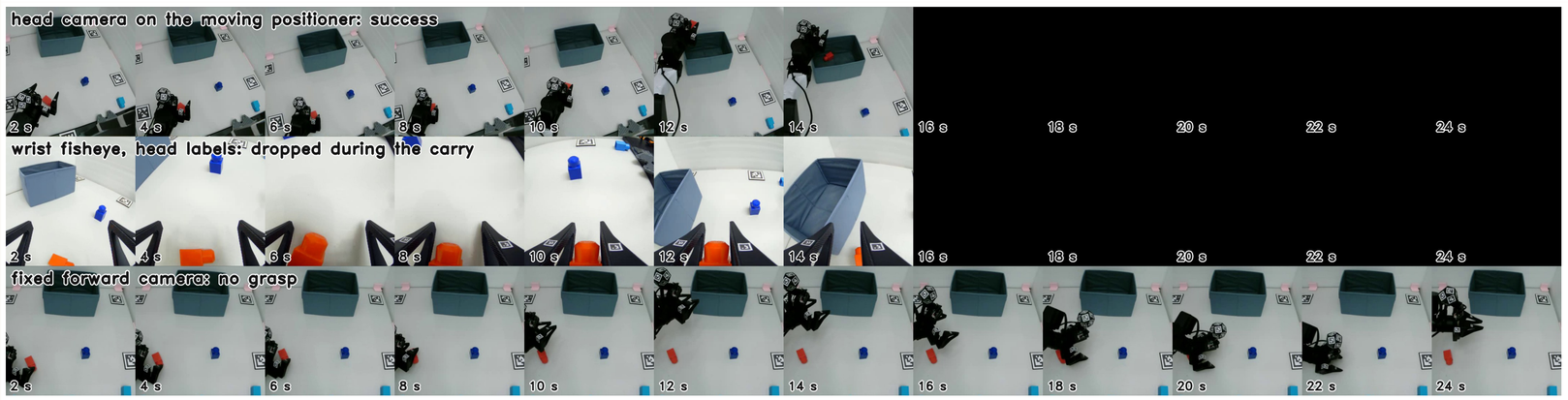
Where each view fails

failures of 25	no grasp	**dropped**	success
head camera	5	**0**	20
wrist fisheye (UMI)	6	**4**	15
wrist fisheye, head labels	3	**8**	14
fixed camera	12	**1**	12

Dropped = during the carry or at the bin. All four co-trained; offline loss within 0.011 of each other.

Wrist-camera policies **drop the block mid-carry**, every time while the fisheye looked away from the bin. The head camera and the fixed camera keep the bin in frame and almost never drop; they fail by not grasping. Caveat: the teleoperator watched the head-camera preview while demonstrating.

One placement, seen by each policy



Placement 23, each policy's own frames every 2 s on one clock, black once its episode ended. Head camera: gripper and bin in one frame through the carry, done in 16 s. Wrist fisheye: only block and wall after the grasp; drops the block before reaching the bin. Fixed camera: sees the whole scene but never closes the gripper (41 s, cut here).

It is the view, not the labels

- Same wrist view, labels from the fixed camera or from the head camera: **15/25 vs 14/25**, no change.
- Same head labels, wrist view or head view: **14/25 → 20/25**, and the drops disappear.

Limits and next

One task, one collector, one tile layout; teleoperation watched the head-camera view. Glove-only policies do not transfer: co-training is required. Next: bimanual, tile-free in the wild.